Script-Controlled ACL – Restrict Record Access Based on FieldValue

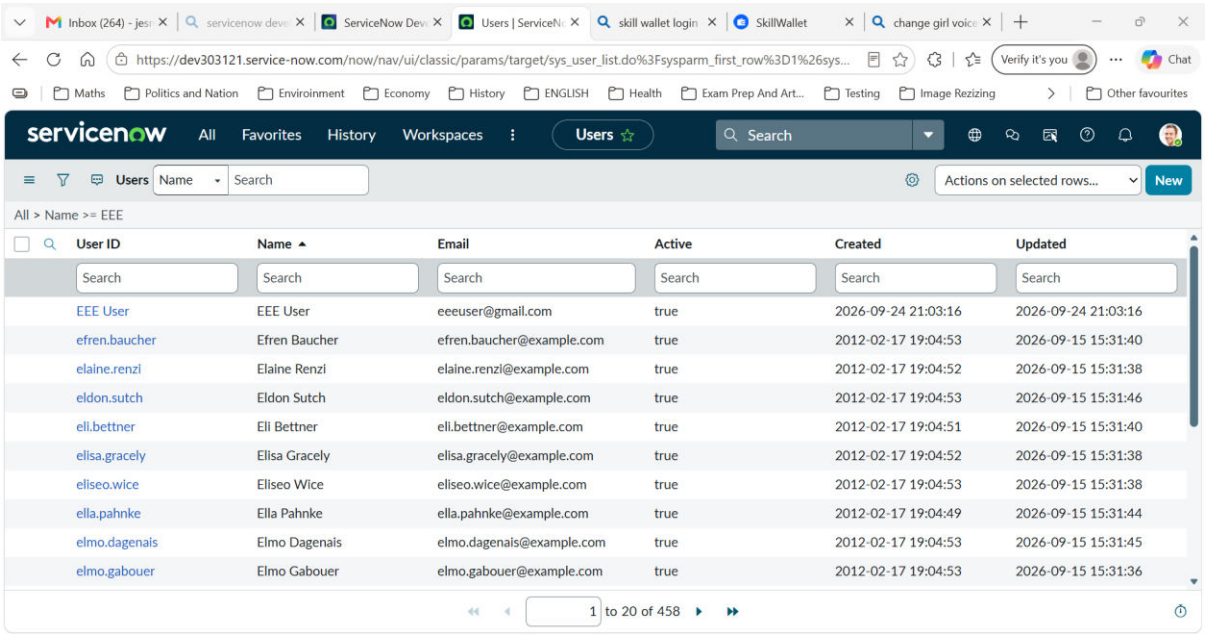
Project Overview:

A Script-Controlled ACL restricts access to records by using a custom script. The script checks specific field values and user conditions before granting access. Users can access the record only when the ACL script evaluates to true**.**

Milestone 1: Creation of Users and Roles

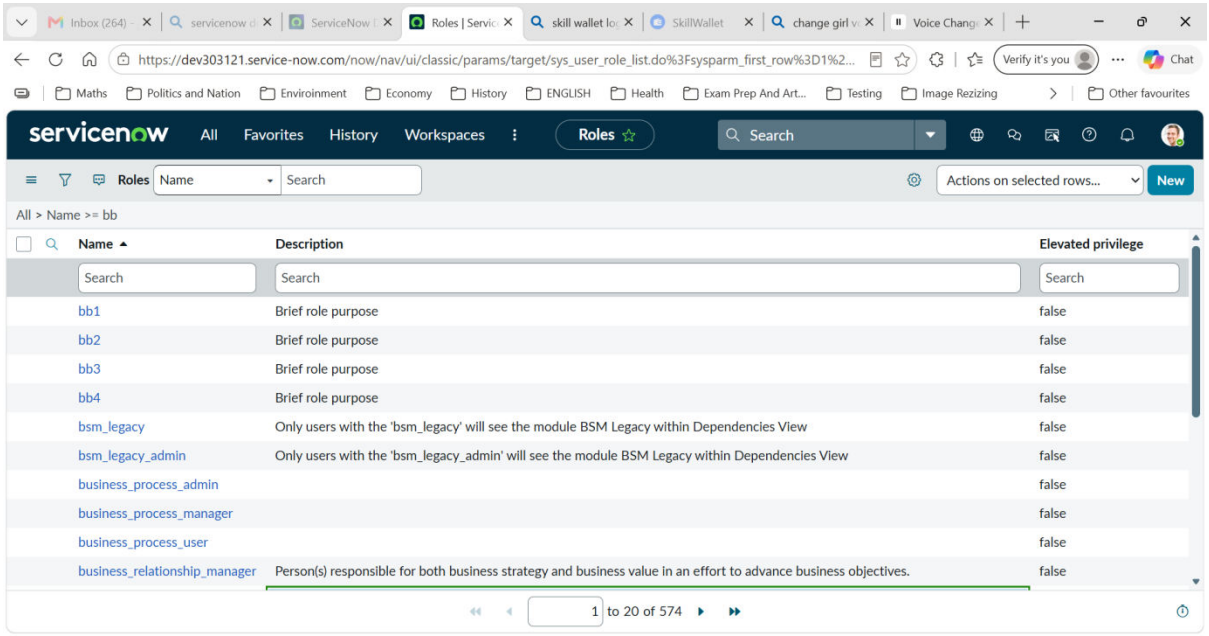
Users Creation

1. Log in to the ServiceNow instance with admin access.
2. Create a test user:
3. Navigate to User Administration → Users
4. Click New
5. User ID: EEE User
6. First name: EEE
7. Last name: User
8. Email: eeeuser@gmail.com
9. Save the record



Roles Creation

1. Navigate to User Administration → Roles
2. Click New
3. Name: Enter role name as bb1
4. Description: Brief role purpose
5. Click Submit
6. Create other roles also like bb2,bb3,bb4.(follow the same procedure)
7. Assign all the roles to the EEE User



Milestone 2: Tables Creation

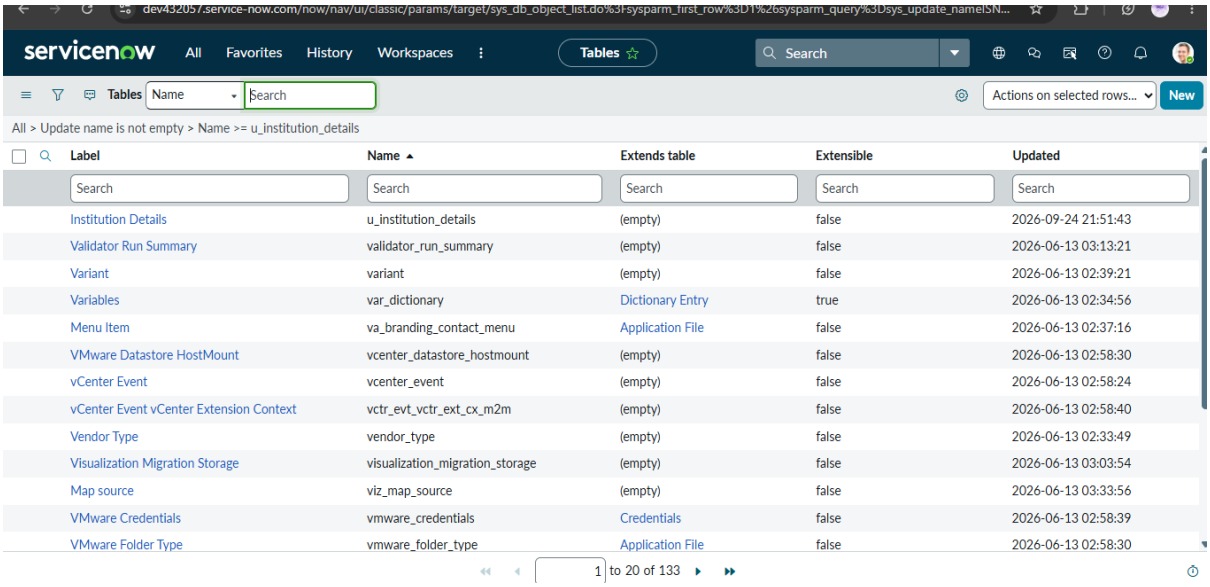
Creation of Table

1. In the Application Navigator, search for System Definition → Tables.
2. Click New and create a custom table with the following details
3. Label: Institution Details
4. Name: u_institution_details
5. Extends for : false
6. Save or Submit the form

   In the columns tab, user Insert a New row method to create the mentioned fields as shown below picture
7. The following fields to the u_institution_details table

8. Student Roll Number– Auto Number
9. Student Name – Reference- User
10. Faculty Name – Reference- User
11. Branch – Choice (ECE, EEE, CSE)
12. Email– String
13. Phone Number– String
14. Description– Multi String
15. Save the Table

    Click on the Show List in the Related Links
16. Create multiple records in the Student Records table with different branch values (ECE, EEE, CSE).

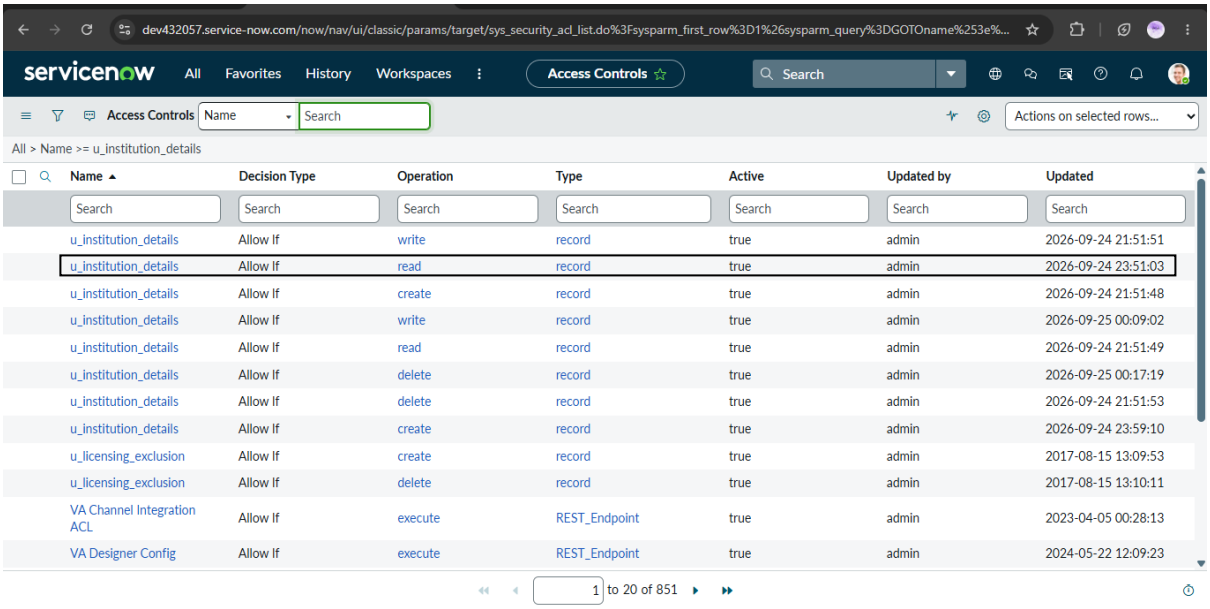


Milestone 3: Creation of Access Control List-READ

ACL Configuration

1. In the Application Navigator, search for System Security → Access Control (ACL).
2. Elevate “Security_admin” role
3. Click New to create a new ACL.
4. Enter the following details
5. Type: record
6. Operation: read
7. Name: u_institution_details
8. Active: true
9. Advanced: true

10. In the Requires role related list, add: bb1 (custom role)
11. Add Data condition as Branch is EEE
12. In the Script field, add the following script
13. Save the ACL form
14. Impersonate EEE users and open the Student Records list.
15. Copy the table name u_institution_details.list/LIST and paste in the All menu.
16. Impersonate Admin and open the Student Records list.
17. Verify : User with bb1 role can view only EEE branch records
18. User without the role cannot view any records
19. Admin user can view all records regardless of branch.



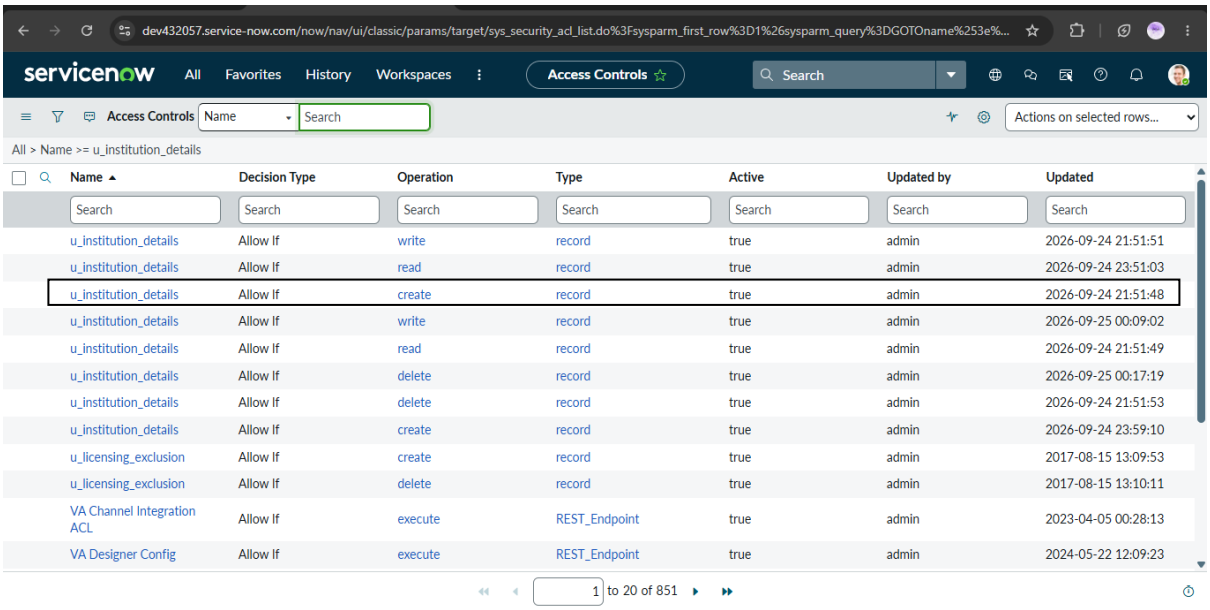
Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_institution_details	Allow If	write	record	true	admin	2026-09-24 21:51:51
u_institution_details	Allow If	read	record	true	admin	2026-09-24 23:51:03
u_institution_details	Allow If	create	record	true	admin	2026-09-24 21:51:48
u_institution_details	Allow If	write	record	true	admin	2026-09-25 00:09:02
u_institution_details	Allow If	read	record	true	admin	2026-09-24 21:51:49
u_institution_details	Allow If	delete	record	true	admin	2026-09-25 00:17:19
u_institution_details	Allow If	delete	record	true	admin	2026-09-24 21:51:53
u_institution_details	Allow If	create	record	true	admin	2026-09-24 23:59:10
u_licensing_exclusion	Allow If	create	record	true	admin	2017-08-15 13:09:53
u_licensing_exclusion	Allow If	delete	record	true	admin	2017-08-15 13:10:11
VA Channel Integration ACL	Allow If	execute	REST_Endpoint	true	admin	2023-04-05 00:28:13
VA Designer Config	Allow If	execute	REST_Endpoint	true	admin	2024-05-22 12:09:23

Milestone 4: Creation of Access Control List-CREATE

Creation of ACL-Create

1. Click New to create a new ACL(Create)
2. Enter the following details
3. Type: record
4. Operation: create
5. Name: u_institution_details
6. Active: true
7. In the Requires role related list, add: bb2 (custom role)
8. No need to add any Data condition
9. Save the ACL Record

10. Verify :
11. User with bb1 &bb2 roles can view only EEE branch records & New Button.

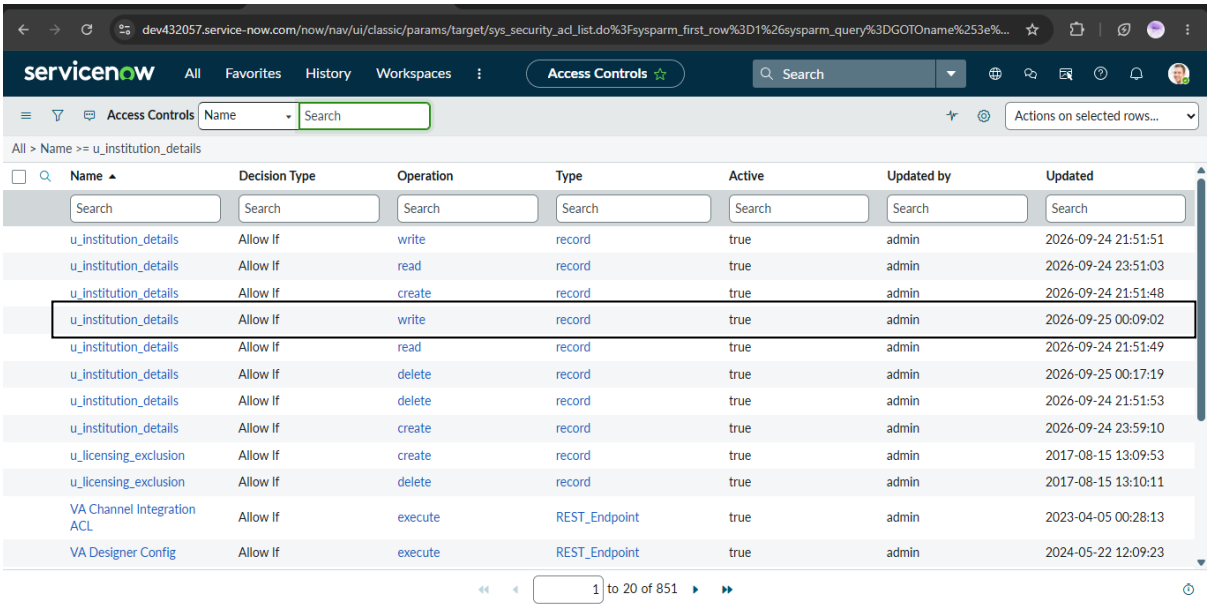


Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_institution_details	Allow If	write	record	true	admin	2026-09-24 21:51:51
u_institution_details	Allow If	read	record	true	admin	2026-09-24 23:51:03
u_institution_details	Allow If	create	record	true	admin	2026-09-24 21:51:48
u_institution_details	Allow If	write	record	true	admin	2026-09-25 00:09:02
u_institution_details	Allow If	read	record	true	admin	2026-09-24 21:51:49
u_institution_details	Allow If	delete	record	true	admin	2026-09-25 00:17:19
u_institution_details	Allow If	delete	record	true	admin	2026-09-24 21:51:53
u_institution_details	Allow If	create	record	true	admin	2026-09-24 23:59:10
u_licensing_exclusion	Allow If	create	record	true	admin	2017-08-15 13:09:53
u_licensing_exclusion	Allow If	delete	record	true	admin	2017-08-15 13:10:11
VA Channel Integration ACL	Allow If	execute	REST_Endpoint	true	admin	2023-04-05 00:28:13
VA Designer Config	Allow If	execute	REST_Endpoint	true	admin	2024-05-22 12:09:23

Milestone5: Creation of Access Control List-WRITE

Creation of ACL-Write

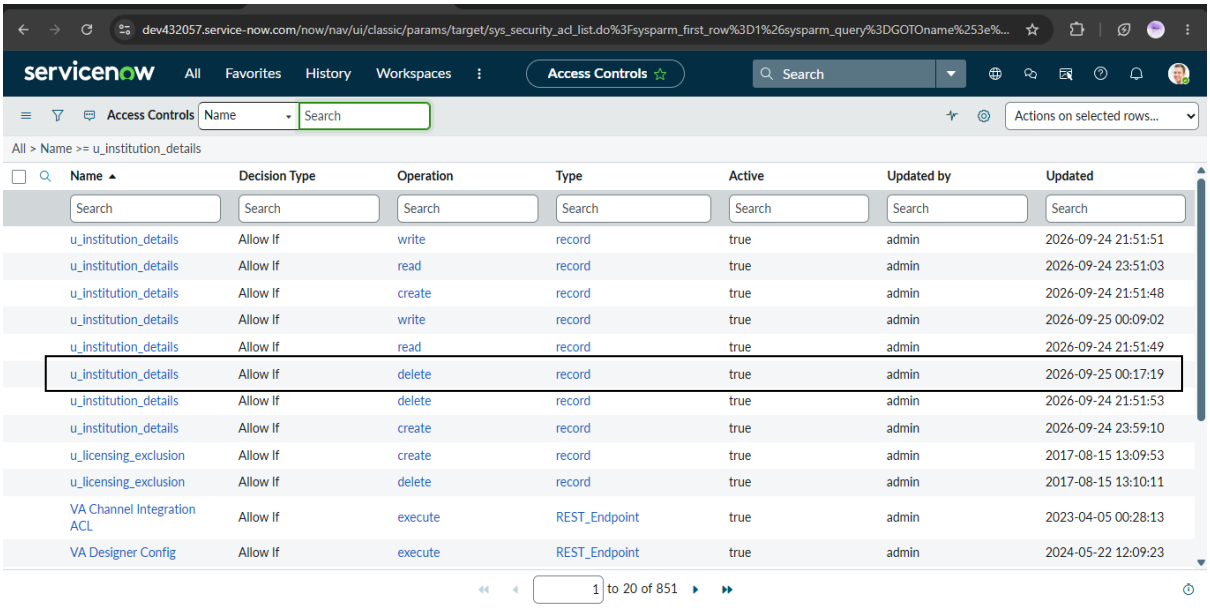
1. Click New to create a new ACL(Write)
2. Enter the following details
3. Type: record
4. Operation: Write
5. Name: u_institution_details
6. Active: true
7. In the Requires role related list, add: bb3 (custom role)
8. No need to add any Data condition
9. Save the ACL Record
10. Verify :
11. User with bb1 &bb2 &bb3 roles can view and edit the EEE branch records along with New Button.



Milestone6: Creation of Access Control List-DELETE

Creation of ACL-DELETE

1. Click New to create a new ACL(Delete)
2. Enter the following details
3. Type: record
4. Operation: Delete
5. Name: u_institution_details
6. Active: true
7. In the Requires role related list, add: bb4 (custom role)
8. No need to add any Data condition
9. Save the ACL Record
10. Verify :
11. Users with bb1 &bb2 &bb3&bb4 roles can see, create and edit along with delete the EEE branch records.



Milestone7: Conclusion

Project Outcome:

Description

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.

